

Technology #14: Acoustic Obstruction Detection for CSF Shunts

Technology Domain: Acoustics + Ultrasound + Signal Detection

Mechanism: Catheter-integrated acoustic transducer that emits ultrasonic pulse and detects reflections from obstruction (debris, tissue, air bubble). Signal classifier identifies obstruction type and severity.

Maturity: ENGINEERING_DEFINITION

Current State: No physical prototype. Computational model and engineering definition exist.

Key Risk: Tissue obstruction acoustic impedance is too similar to CSF for reliable detection (Z ratio < 1.1)

Next Experiment: Acoustic transducer prototypes (various frequencies + geometries)

Buyer Receives: Acoustic obstruction detection concept + mechanism description; Acoustic impedance + reflection + TOA governing equations; Bench test protocols (detection sensitivity, false positives, acoustic output, EMC); External precedent catalog (IVUS, Doppler, AIUM/NEMA UD-2); Critical UNKNOWN disclosure (tissue obstruction sensitivity, optimal frequency, false positive rate); This engineering dossier with failure analysis + build plan

Buyer Must Build: Production transducer design (frequency, geometry, material); Production drive + receive electronics + signal processor; Production classifier (validated in vivo); Manufacturing process for catheter-integrated acoustic detector; Regulatory submission (510(k) with IVUS predicate, or De Novo); Clinical validation (sensitivity + specificity vs imaging)

Transfer Posture: Sponsored Validation

Target Buyer: Neuroshunt manufacturer; ultrasound device company

CONFIDENTIAL. See full engineering dossier for details.